Predicting Cannabis Usage Behaviour Using Machine Learning and Personality Traits

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Github link:https: //github.com/Binnari2001/drugs/upload

Introduction & Motivation

Psychological, behavioral, and demographic factors affect how people use drugs and alcohol.

Prior studies indicate that personality characteristics, including impulsivity, sensation-seeking, and emotional stability, may affect patterns of drug consumption.

The existence of behavioral datasets facilitates the application of machine learning methodologies to examine intricate correlations between personality traits and substance consumption.

Knowing what makes people use cannabis can help with behavioral and public health research.

Machine learning models can find patterns in demographic and personality data that might be linked to how people use drugs.

Research question & Objective

Is it possible for machine learning models to accurately predict how much cannabis a person will use based on their personality and demographic traits?

- Look at the Drug Consumption dataset and find demographic and personality traits that are linked to using cannabis.
- Use SMOTE to change the drug categories and fix the class imbalance in the dataset.
- Make and improve machine learning models that can predict how people will use cannabis.
- Use classification metrics to look at and compare how well the models work.

Literature Review

- Past studies have shown that personality traits and demographic factors are important for predicting how people will use drugs. Machine learning can look at behavioral data and find patterns that can help predict drug use and the risk of addiction (Nath, Kilam, and Swetapadma, 2017).

- Researchers have used the Drug Consumption dataset to show that personality traits like neuroticism, extraversion, openness, agreeableness, and conscientiousness can help predict how people use drugs. These features can be used by machine learning models to accurately sort drug use behavior (Adinugroho, Sari, and Hidayat, 2019).

- Researchers have also used different machine learning algorithms, like Support Vector Machines, Random Forest, and Gradient Boosting, to compare drug use patterns. These studies demonstrate that model performance is contingent upon the dataset, feature selection, and evaluation metrics employed (Zhang, 2025).

- Recent research has shown even more how well ensemble learning methods work for analyzing complicated addiction-related datasets. Models like Random Forest have shown to be very good at predicting drug-related outcomes using behavioral and clinical variables (Tapia-Galisteo et al., 2020).

Dataset Overview

1. Source of the dataset
The study uses the Drug Consumption dataset from the UCI Machine Learning Repository. This dataset has information about people's behavior, personality, and demographics that are related to drug use.

2. Size of the dataset
With 1885 participants and 32 variables, the dataset has enough information to look for patterns in how people use drugs.

3. Types of Features
The dataset has demographic information like age, gender, education, country, and ethnicity, as well as personality traits based on the Big Five model and behavioral traits like being impulsive and seeking out new experiences.

4. The variable you want to target
Cannabis use is the main thing this study tries to predict. It uses this information to teach machine learning models how to sort people based on how likely they are to use cannabis.

Ethical Considerations

1. Keeping data private and anonymous
There is no personally identifiable information in the Drug Consumption dataset, so the identities of the people who took part are kept private and safe.

2. How to Use Secondary Data in a Moral Way
The dataset came from the UCI Machine Learning Repository, which offers publicly available datasets for academic research. This makes sure that the data can be used for analysis in an ethical way.

3. Using Predictive Models in a Responsible Way
The machine learning models created in this project are only for research and analysis. Predictions should not be used to label or shame people.

4. Being aware of ethics in data science
This study follows ethical data science by making sure that the data is used responsibly, that the results are carefully interpreted, and that the data is open to everyone.

Data Preprocessing

1. Cleaning up data: We looked through the dataset to see if there were any missing values or duplicate records. The analysis showed that the dataset was clean, which meant that the data could be used to train models with confidence.

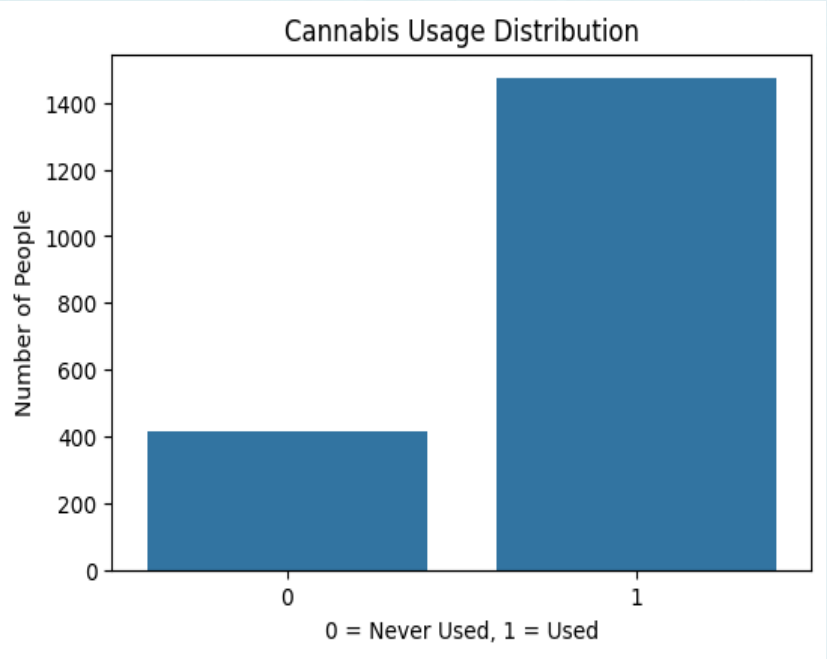
2. Changes to Drug Categories: We turned the original drug use levels (CL0–CL6) into numbers and then into a binary classification variable, where 0 means not using drugs and 1 means using drugs.

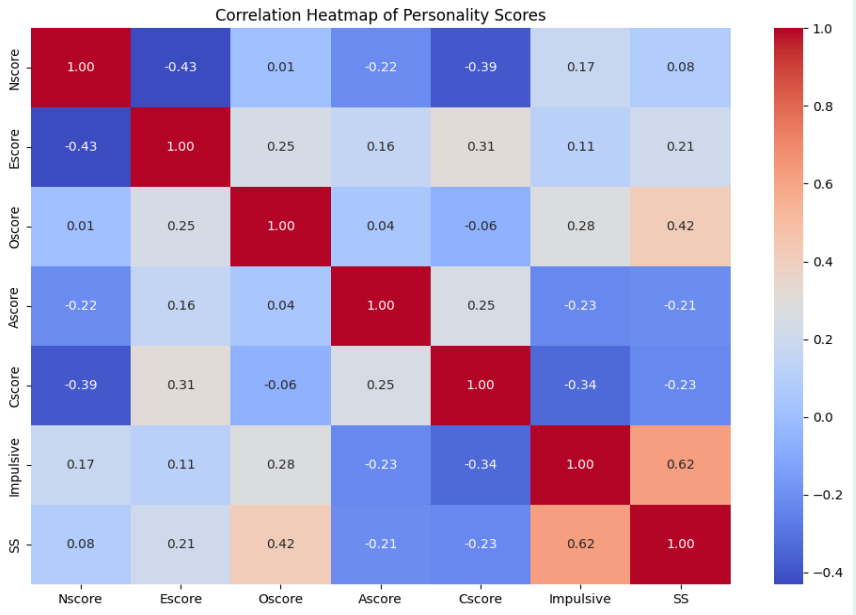
3. Choosing features: We chose relevant demographic, personality, and behavioral variables as input features, and the cannabis use variable as the target variable for prediction.

4. Dealing with Class Imbalance: There were more cannabis users than non-users in the dataset. We used the SMOTE (Synthetic Minority Oversampling Technique) to make the training data more even and make the model work better.

Exploratory Data Analysis (EDA)

- An exploratory analysis revealed that the dataset comprises 1885 observations and 32 variables, encompassing demographic, personality, and behavioral characteristics associated with drug consumption.
- We looked for missing values and duplicate records in the dataset. The analysis showed that there were no missing values or duplicates, which meant that the dataset was clean and could be used for machine learning.
- EDA showed that the cannabis usage variable was unbalanced, with a lot more users than non-users. This imbalance meant that SMOTE had to be used during preprocessing to make the training data more even.
- We used statistics and graphs to look at the links between personality traits, demographic features, and cannabis use. This helped us find the right predictors for the machine learning models.





Methodology and Machine Learning Models

1. Getting the data ready and balancing the classes: We changed the drug use categories in the dataset into a binary cannabis use variable before using it. The dataset had an uneven number of users and non-users, so the SMOTE (Synthetic Minority Oversampling Technique) was used to balance the training data and help the model learn better.

2. Used Machine Learning Models: We used three supervised machine learning classification algorithms: Random Forest, Support Vector Machine (SVM), and XGBoost. We chose these models because they are commonly used for classification tasks and do a good job of finding complex relationships in behavioral datasets.

3. Training the model and finding the best hyperparameters: To test how well the model worked, the dataset was split into training and testing sets. We used RandomizedSearchCV with cross-validation to find the best hyperparameter settings for each algorithm and to improve the model.

4. Evaluating and Comparing Models: We used classification metrics like accuracy, F1-score, and confusion matrix to rate the models. These metrics let you compare model performance in a thorough way and find the model that works best for predicting cannabis use.

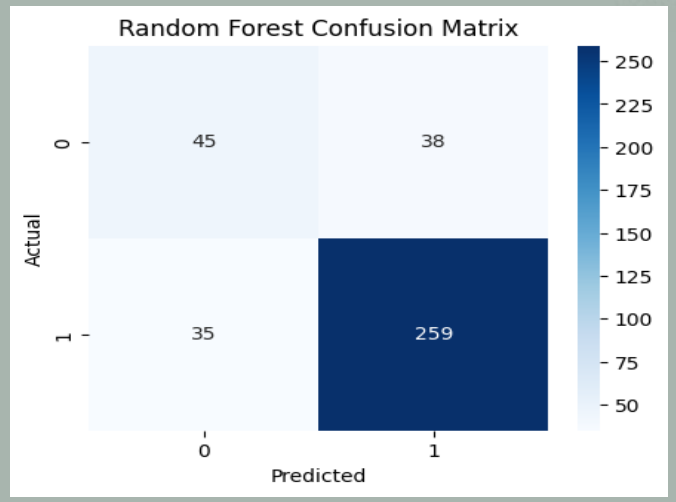
Model Results and Performance

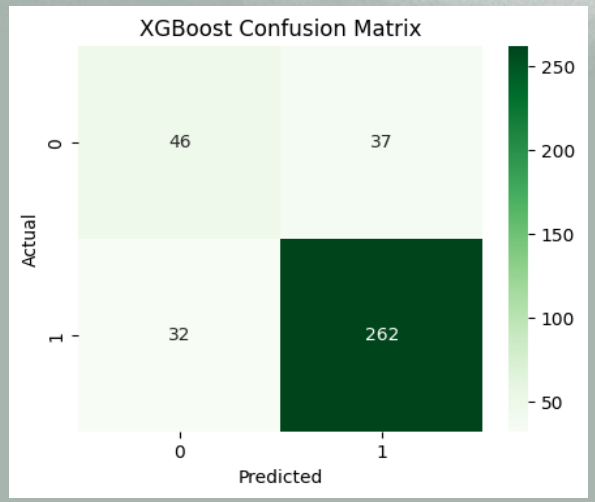
- 1. How to Evaluate: I used accuracy, F1-score, and confusion matrix to check how well the machine learning models worked. These are all good ways to check how well a model does at classification.
- 2. Comparing Different Machine Learning Models: We used three models: Random Forest, Support Vector Machine (SVM), and XGBoost. To make sure that the evaluation was fair, each model was trained on the same set of preprocessing steps and a balanced dataset.
- 3. Important Results:The findings suggest that personality traits, demographic factors, and behavioral characteristics can significantly enhance the prediction of cannabis usage behavior.
- 4. The best model performance: XGBoost had the best overall performance of the algorithms that were tested. This suggests that ensemble-based methods work well for analyzing complex behavioral datasets.

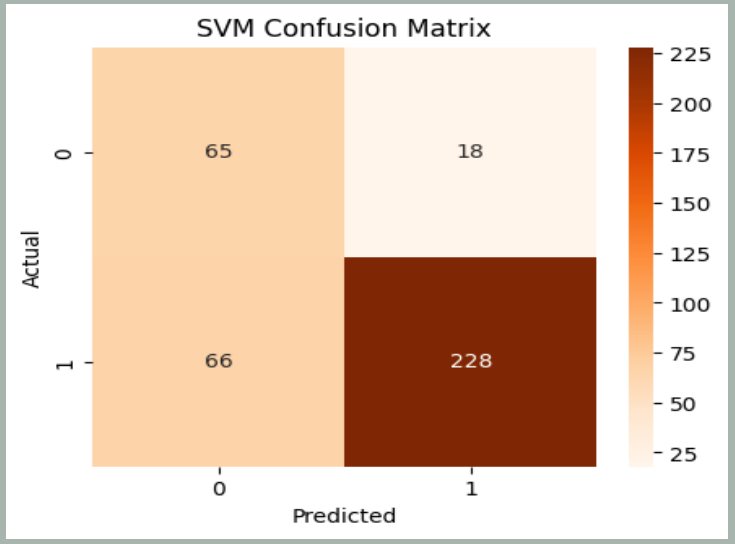
Model	Accuracy	Precision	Recall	F1-Score
Random Forest	0.806	0.8	0.81	0.81
SVM	0.777	0.83	0.78	0.79
XGBoost	0.817	0.81	0.82	0.81

Confusion Matrix

- Random Forest confusion matrix shows strong detection of cannabis users. ROC curve indicates good classification ability between users and non-users.
- SVM confusion matrix shows better identification of non-users but more user misclassification. ROC curve indicates moderate discrimination between classes.
- XGBoost confusion matrix shows the best balance identifying users and non-users. ROC curve demonstrates strong predictive performance and class separation.



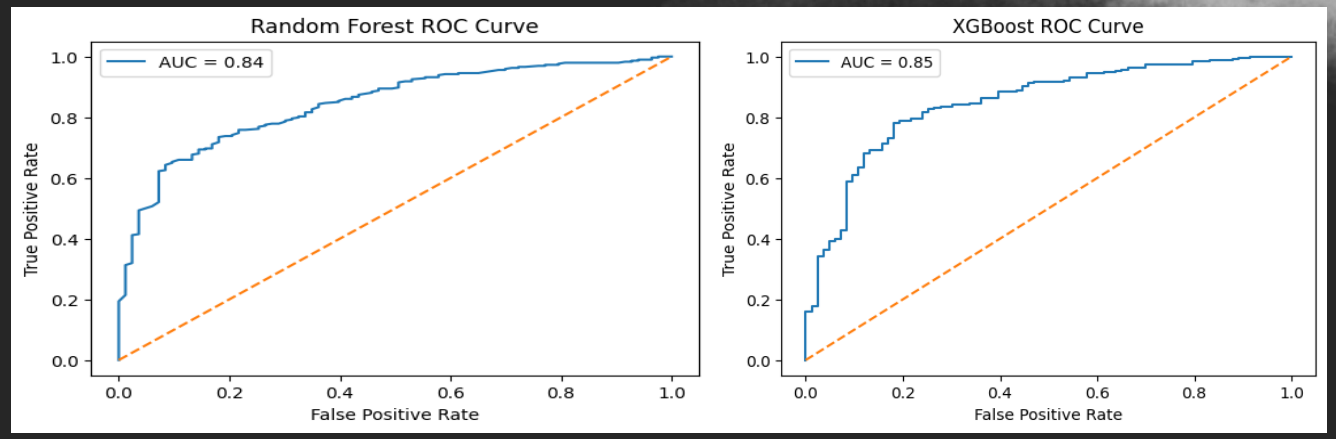


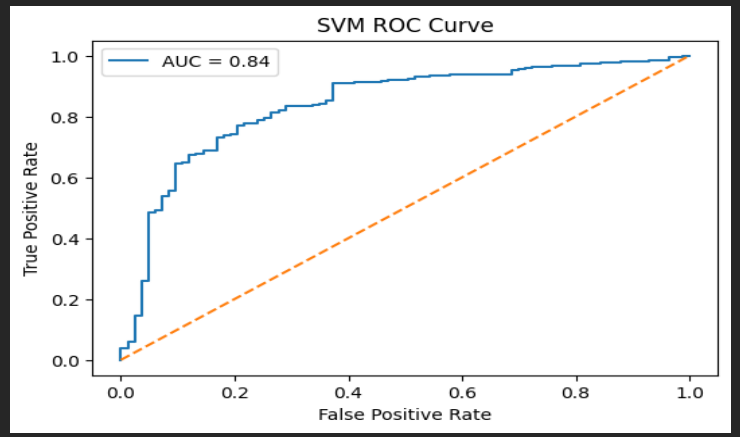


Model	Correctly Predicted Non-Users (0)	Correctly Predicted Cannabis Users (1)
Random Forest	45	259
SVM	65	228
XGBoost	46	262

ROC Curve

- The ROC curve for Random Forest shows that it can tell the difference between people who use cannabis and those who don't, which means it can classify things well.(AUC = 0.84))
- The ROC curve for SVM shows that the classes are somewhat separated, which means that the model works well but doesn't do as well as others at telling them apart.(AUC =0.8)
- The ROC curve for XGBoost shows the best ability to tell the difference between cannabis users and non-users because it has the strongest class separation.(AUC = 0.84)





XGBoost got the highest AUC score (0.85), which means it was the best at telling the difference between people who use cannabis and those who don't.

Hyperparameter Optimisation Results

- **Method for tuning hyperparameters:** We used RandomizedSearchCV with cross-validation to find the best parameter combinations for the Random Forest, SVM, and XGBoost models.
- **How well the model works after being optimized:** The models got the following accuracies after tuning:
  - **Random Forest: 0.793**
  - **SVM: 0.790**
  - **0.820 for XGBoost**
- **The model that worked best:** The tuned XGBoost model had the best accuracy (82%) and found the most cannabis users, which shows that it was the best at making predictions.

Model	Accuracy	Correct Non-Users (0)	Correct Cannabis Users (1)
Tuned Random Forest	0.793	41	258
Tuned SVM	0.79	48	250
Tuned XGBoost	0.82	43	266

Summary and conclusions

- This study used machine learning to predict cannabis use based on demographic, personality, and behavioral data from the Drug Consumption dataset.
- The study found that personality and behavior traits can help predict how people use cannabis.
- Of all the models tested, XGBoost had the best predictive accuracy (0.82) and was best at telling the difference between cannabis users and non-users.
- In general, the results show that machine learning models can find patterns in behavioral datasets and help predict how people will use cannabis.

References

P. Nath, S. Kilam and A. Swetapadma, "A machine learning approach to predict volatile substance abuse for drug risk analysis," *2017 Third International Conference on Research in Computational Intelligence and Communication Networks (ICRCICN)*, Kolkata, India, 2017, pp. 255-258, doi: 10.1109/ICRCICN.2017.8234516.
URL: https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8234516&isnumber=8234465

S. Adinugroho, Y. A. Sari and N. Hidayat, "Drug usage duration classification using Extreme Learning Machine based on personality features," *2019 International Conference on Sustainable Information Engineering and Technology (SIET)*, Lombok, Indonesia, 2019, pp. 33-37, doi: 10.1109/SIET48054.2019.8986131.
URL: https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8986131&isnumber=8985986

Z. Yi, "Constructing a Drug Consumption Prediction Model Based on Machine Learning Strategies," *2025 International Symposium on Intelligent Robotics and Systems (ISoIRS)*, Chengdu, China, 2025, pp. 1-6, doi: 10.1109/ISoIRS65690.2025.11167906.
URL: https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=11167906&isnumber=11167892

J. Tapia-Galisteo *et al*., "Prediction of Cocaine Inpatient Treatment Success Using Machine Learning on High-Dimensional Heterogeneous Data," in *IEEE Access*, vol. 8, pp. 218936-218953, 2020, doi: 10.1109/ACCESS.2020.3041895.
URL: https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9276472&isnumber=8948470

THANK YOU!